

SME SettlementChain

Hybrid Blockchain for Invoice Settlement, Risk Analytics, and Decentralized Financial Workflows

Overview

SME SettlementChain is a hybrid blockchain system designed to model real-world financial workflows for small and medium enterprises (SMEs).

Unlike traditional blockchains that only handle currency transfers, this platform integrates a decentralized ledger with a full business logic layer, enabling structured financial interactions such as invoice issuance, validation, financing, and settlement.

The system simulates a complete financial network where buyers, suppliers, and lenders interact transparently without relying on centralized intermediaries.

What Makes This Different

This system combines:

- Blockchain (PoW, Merkle trees, validation)
- Distributed Systems (P2P networking, node sync)
- Backend Engineering (REST API, transaction flow)
- Fintech Logic (invoice lifecycle, settlement, liquidity)
- Compliance Simulation (KYC, regulated participation)

System Components

Java Node (java-node/)

- Proof-of-Work blockchain implementation
- P2P networking
- Transaction validation and mempool
- Invoice settlement state machine
- REST API
- Snapshot export

Python Analytics (python-analytics/)

- Risk scoring
- Financial analytics
- Reporting

Financial Workflow

1. Supplier issues invoice
2. Buyer validates

3. Lender provides liquidity (simulated)

4. Settlement occurs on-chain

Tech Stack

- Java
- Python
- REST APIs
- JSON-over-TCP
- Ed25519 cryptography

Quickstart

Run node:

cd java-node

./gradlew run --args="--port 8080 --p2pPort 9000 --dataDir data/node1"

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